Practice Quiz: Lecture 3 – Part 1

Topics: Signal Representation, Oscilloscopes, Sampling Theorem & Aliasing, Signal Bandwidth, Bit Duration, Signal Levels (L), and Signal Rate vs. Bit Rate

Student Name: _____ **Student ID:** _____ **Score:** _____ / 20

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1. Which statement best describes the fundamental function of an oscilloscope in communication systems?

( ) A. It computes and displays the IP address and packet routing path across an interface.

( ) B. It displays the amplitude variations of an electrical signal as a function of time.

( ) C. It converts digital bitstreams directly into continuous acoustic analog messages.

( ) D. It measures exclusively the spectral bandwidth across the frequency domain.

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2. In analog-to-digital signal conversion, what is the precise definition of "sampling" a signal?

( ) A. Eliminating all high-frequency harmonics using an active low-pass filter.

( ) B. Multiplying the signal amplitude by a constant gain factor prior to transmission.

( ) C. Measuring and recording the instantaneous amplitude value of the continuous signal at discrete, uniform points in time.

( ) D. Modulating multiple baseband channels into a single high-frequency carrier wave.

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3. What undesirable phenomenon occurs when a continuous signal is sampled at a rate strictly lower than its Nyquist rate ($f_s < 2f_{\max}$)?

( ) A. The reconstructed signal amplitude increases exponentially while frequency is preserved.

( ) B. High-frequency components fold back and appear as false lower-frequency components (aliasing effect).

( ) C. The signal bandwidth automatically expands to match the sampling frequency.

( ) D. The bit interval drops to zero, producing immediate transmission attenuation.

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4. A standard voice signal contains significant frequency components ranging from $300\ \text{Hz}$ to $3400\ \text{Hz}$. According to the Nyquist theorem, what is the minimum sampling rate required to avoid aliasing?

Answer: _____

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5. A pure sinusoidal signal with a frequency of $f = 7\ \text{kHz}$ is sampled at a sampling rate of $f_s = 10\ \text{kHz}$. Due to the aliasing effect, at what apparent frequency will the reconstructed signal appear in the baseband?

Answer: _____

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6. A periodic composite signal consists of four discrete sinusoidal harmonic components at $50\ \text{Hz}$, $150\ \text{Hz}$, $450\ \text{Hz}$, and $1250\ \text{Hz}$. Calculate the total bandwidth of this signal.

Answer: _____

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7. A composite signal has a total bandwidth of $B = 15\ \text{kHz}$. If its lowest frequency component is measured at $f_{\text{low}} = 5\ \text{kHz}$, what is its highest frequency component f_{high}?

Answer: _____

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8. A digital communication system operates at a bit rate of $N = 4000\ \text{bps}$. Calculate the exact duration of a single bit (the bit interval T_b) in microseconds (μs).

Answer: _____

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9. If the duration of one bit on a transmission medium is measured to be $T_b = 50\ \mu\text{s}$, what is the data bit rate (N) in kbps?

Answer: _____

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10. A multi-level digital signaling system uses $L = 16$ distinct discrete voltage levels. How many bits of information are represented by each signal element (symbol)?

Answer: _____

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11. A communication transceiver encodes $r = 5\ \text{bits}$ into each transmitted signal element. What is the minimum number of distinct signal levels (L) required?

Answer: _____

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12. What is the primary communication engineering design objective when optimizing data rate (N) and signal rate (S)?

( ) A. Minimize data rate while maximizing the transmitted baud rate.

( ) B. Increase the data rate while decreasing the required signal rate to conserve bandwidth.

( ) C. Maintain signal rate strictly equal to data rate regardless of available levels.

( ) D. Eliminate all multi-level modulation schemes to force $L = 2$.

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13. A digital transmission scheme carries data where each signal element represents 2 bits ($r = 2$). If the bit rate is $N = 120\ \text{kbps}$ and the case factor is $c = 0.5$, calculate the average signal rate (S) in kbaud.

Answer: _____

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14. A system uses $L = 8$ discrete signal levels. Each signal element carries the maximum possible number of bits. If the bit rate is $N = 240\ \text{kbps}$ and $c = 0.5$, determine the average baud rate (S) in kbaud.

Answer: _____

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15. A modulated transmission operates at an average signal rate of $S = 50\ \text{kbaud}$ with a case factor of $c = 0.5$. If each symbol carries $r = 4\ \text{bits}$, calculate the corresponding data rate (N) in kbps.

Answer: _____

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16. Contrast an oscilloscope with a spectrum analyzer: While an oscilloscope displays a signal in the time domain, a spectrum analyzer displays the signal in which domain?

( ) A. Spatial propagation domain

( ) B. Code division domain

( ) C. Frequency domain (Amplitude vs. Frequency)

( ) D. Quantization noise domain

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17. An audio recording studio processes an acoustic signal containing frequency components up to $22.05\ \text{kHz}$. What is the absolute minimum Nyquist sampling frequency required to capture this signal without aliasing?

Answer: _____

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18. True or False: In any digital transmission system where each signal element carries more than one bit ($r > 1$), the baud rate (signal rate S) is always strictly greater than the bit rate (N).

( ) True

( ) False

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19. A binary signaling system transmits 1 bit per signal element ($r = 1$, $L = 2$) over a channel at $N = 100\ \text{kbps}$. For an average case factor of $c = 0.5$, calculate the average signal rate (S) in kbaud.

Answer: _____

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20. A channel has a strictly constrained bandwidth allowing a maximum average signal rate of $S = 80\ \text{kbaud}$ (with $c = 0.5$). If the required throughput is $N = 480\ \text{kbps}$, calculate the minimum number of discrete signal levels (L) the modulation scheme must support.

Answer: _____

Answer Key & Detailed Explanations (Lecture 3 – Part 1)

Question 1: B

An oscilloscope is a laboratory instrument designed to display electrical signal voltage/amplitude variations across the *time domain*. Frequency domain analysis is performed using a spectrum analyzer.

Question 2: C

Sampling is the process of measuring the amplitude values of a continuous-time analog signal at discrete, periodic time intervals ($T_s = 1/f_s$).

Question 3: B

When $f_s < 2f_{\max}$, undersampling causes spectral overlap (aliasing). Higher-frequency components fold back across the Nyquist frequency ($f_s/2$) and masquerade as lower frequencies upon reconstruction.

Question 4: 6800 Hz (or 6.8 kHz)

By the Nyquist sampling theorem:

$$f_s \geq 2 \times f_{\max} = 2 \times 3400 \text{ Hz} = 6800 \text{ Hz} = 6.8 \text{ kHz}$$

Question 5: 3 kHz

The folding (Nyquist) frequency is $f_N = f_s/2 = 10/2 = 5$ kHz. Because $f = 7 \text{ kHz} > 5 \text{ kHz}$, the signal aliases at:

$$f_{\text{alias}} = |f_s - f| = |10 \text{ kHz} - 7 \text{ kHz}| = 3 \text{ kHz}$$

Question 6: 1200 Hz

Bandwidth (B) is the difference between the highest and lowest frequencies:

$$B = f_{\text{high}} - f_{\text{low}} = 1250 \text{ Hz} - 50 \text{ Hz} = 1200 \text{ Hz}$$

Question 7: 20 kHz

$$B = f_{\text{high}} - f_{\text{low}} \implies f_{\text{high}} = f_{\text{low}} + B = 5 \text{ kHz} + 15 \text{ kHz} = 20 \text{ kHz}$$

Question 8: 250 μs

The bit interval T_b is the reciprocal of the bit rate N:

$$T_b = \frac{1}{N} = \frac{1}{4000 \text{ bps}} = 0.00025 \text{ s} = 250 \ \mu\text{s}$$

Question 9: 20 kbps

$$N = \frac{1}{T_b} = \frac{1}{50 \times 10^{-6} \text{ s}} = 20,000 \text{ bps} = 20 \text{ kbps}$$

Question 10: 4 bits

The number of data bits per symbol r is:

$$r = \log_2(L) = \log_2(16) = 4 \text{ bits/symbol}$$

Question 11: 32 levels

$$L = 2^r = 2^5 = 32 \text{ distinct signal levels}$$

Question 12: B

A primary goal in data communications is to maximize data throughput (N) while minimizing signal transitions (S), because transmission bandwidth is proportional to signal rate.

Question 13: 30 kbaud

Using the signal rate formula:

$$S = c \times \frac{N}{r} = 0.5 \times \frac{120 \text{ kbps}}{2} = 30 \text{ kbaud}$$

Question 14: 40 kbaud

With $L = 8$, $r = \log_2(8) = 3$ bits/symbol. Therefore:

$$S = c \times \frac{N}{r} = 0.5 \times \frac{240 \text{ kbps}}{3} = 0.5 \times 80 = 40 \text{ kbaud}$$

Question 15: 400 kbps

Rearranging $S = c \times \frac{N}{r}$:

$$N = \frac{S \times r}{c} = \frac{50 \text{ kbaud} \times 4}{0.5} = \frac{200}{0.5} = 400 \text{ kbps}$$

Question 16: C

An oscilloscope maps Amplitude vs. Time (time domain); a spectrum analyzer maps Amplitude/Power vs. Frequency (frequency domain).

Question 17: 44.1 kHz

$$f_s \geq 2 \times f_{\max} = 2 \times 22.05 \text{ kHz} = 44.1 \text{ kHz}$$

(This is the exact reason standard compact disc audio is sampled at 44.1 kHz).

Question 18: False

Because $S = c \times \frac{N}{r}$, when $r > 1$ each symbol represents multiple bits, making the baud rate strictly *less* than the bit rate ($S < N$).

Question 19: 50 kbaud

$$S = c \times \frac{N}{r} = 0.5 \times \frac{100 \text{ kbps}}{1} = 50 \text{ kbaud}$$

Question 20: $L = 8 \text{ levels}$

First find the required bits per symbol r:

$$S = c \times \frac{N}{r} \implies 80 = 0.5 \times \frac{480}{r} \implies 80 = \frac{240}{r} \implies r = \frac{240}{80} = 3 \text{ bits/symbol}$$

The required number of signal levels is:

$$L = 2^r = 2^3 = 8 \text{ levels}$$
